
Reasoning Depth Erodes the Premise Verification That Post Training Builds

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Abstract

Whether a language model can tell that a problem does not determine an answer has been measured on released models, which cannot say when the ability arrives in training or what takes it away. Instella publishes intermediate checkpoints spanning pretraining and post training together with its full data mixture, so both questions become measurable. A premise deletion probe removes exactly one premise the reference solution consumes, giving 3179 variants over 1779 parent problems, and every abstention rate is read against an answerable control on the untouched parents, where declining is an error. Verification is close to absent before post training and is built by it: discrimination runs +1.84, +10.10, +38.08 and +52.63 points across stage one, stage two, supervised fine tuning and instruct, while wrong refusals stay below 3.5 percent throughout, and a reworded licence preserves the ladder. An independently built MATH probe reproduces it, and three instruct checkpoints from other families all discriminate. Reasoning depth then erodes what training builds. Abstention on underdetermined problems falls 5.22 points per additional calculator step while the answerable control stays flat at -0.11, a difference of -5.11 on $[-6.36, -3.82]$, and the decay steepens as the checkpoint improves. Under the answer forcing prompt used for benchmark scoring the strongest checkpoint still answers 97.6 percent of provably underdetermined items. Controlled injection at a fixed token budget drives verbatim reproduction to 0.946 and returns the memorised calculation chain, at +5.44 on $[+1.46, +9.38]$, without returning the answer, at +1.52 on $[-2.12, +5.21]$, so what depth erodes is not recall of training text.

1 Introduction

A model that answers a grade school arithmetic problem has been given premises that determine an answer. Whether it checked that they do is invisible to the score, because every item on a benchmark is answerable. Probing that directly requires items that are not, and a growing body of work now supplies them. QuestBench casts underspecified reasoning tasks as constraint satisfaction problems with a missing variable [Li et al., 2025a], AbstentionBench assembles unanswerable questions across domains and reports that reasoning fine tuning degrades abstention [Kirichenko et al., 2025], PCBench evaluates premise critique across fifteen models and finds that most detect a flawed premise only when explicitly prompted to look [Li et al., 2025b], and ReliableMath does the same for unsolvable mathematics [Xue et al., 2025]. That models often fail to verify their premises, and do better when told they may decline, is therefore established rather than at issue here.

What that work cannot ask is developmental. Every model it evaluates is a finished artifact, so the capability can be measured but not located. Answering that needs published intermediate checkpoints and the mixture each one consumed. OLMo 2 releases thousands of checkpoints [Team OLMo et al., 2024]; Instella [Liu et al., 2025] releases four spanning pretraining through instruction tuning, and pairs them with a synthetic mathematics corpus that enters at a named stage boundary, which is what

Table 1: Declining with its control, under a prompt that names abstention as an option. Pruned items are underdetermined so declining is correct; parent problems are answerable so declining is an error. The last two columns give balanced accuracy of the decline decision, where 0.500 is chance, measured on the GSM8K probe and on an independently built MATH probe of 818 parents. Cluster bootstrap over parent problems, 4000 draws, seed 6198.

Checkpoint	GSM8K probe				Decision quality	
	Declines, und.	Declines, ans.	Discrimination	95% CI	GSM8K	MATH
Stage 1	0.043	0.025	+1.84	[+0.97, +2.70]	0.509	0.516
Instella 3B	0.110	0.008	+10.10	[+8.94, +11.30]	0.550	0.557
SFT	0.416	0.035	+38.08	[+36.06, +40.02]	0.690	0.611
Instruct	0.538	0.011	+ 52.63	[+50.77, +54.57]	0.763	0.675

38 makes exposure here measurable rather than estimated. This paper asks when premise verification
39 appears, and what it is lost to.

40 Both answers turn out to concern reasoning rather than compliance. Verification is close to absent in
41 the pretrained checkpoints and is manufactured by post training, rising from +1.84 to +52.63 points
42 of discrimination against an answerable control. It then erodes with the depth of the chain the model
43 is tracking, falling 5.22 points per additional reasoning step while the control stays flat, and eroding
44 faster in the checkpoints that verify best. Because the corpus is published, the erosion can also be
45 separated from memorisation: continued pretraining that drives verbatim reproduction of injected
46 documents to 0.946 returns the memorised calculation chain without returning the answer, so what
47 fails at depth is the check, not the recall.

48 2 Setup

49 **Exposure:** Containment is the fraction of an item’s 13-grams appearing in a single corpus document,
50 taking the maximum over documents. Calibration against the GSM8K test split, which the released
51 corpus does not contain, is in Appendix B.

52 **The probe:** Each item removes one premise the reference solution consumes, verified by requiring
53 the annotated calculator chain to depend on it. The result is 3179 variants over 1779 parent problems,
54 split 1591 train and 1588 test. Every item is underdetermined by construction, so declining is the
55 correct response and any specific number is wrong.

56 **Checkpoints:** Four public Instella checkpoints span the pipeline: stage one, which precedes the
57 synthetic corpus, stage two, supervised fine tuning and instruct. Decoding is greedy and held fixed;
58 base checkpoints are prompted with four exemplars and no chat template, since a base model loops
59 without them.

60 **Inference:** Intervals are cluster bootstraps over parent problems with 4000 draws at seed 6198,
61 because several variants can descend from one problem and resampling rows would understate the
62 interval.

63 3 Verification of premise sufficiency

64 3.1 Post training builds the capability

65 Benchmark scoring uses an answer forcing prompt. Under it the instruct checkpoint declines on
66 2.4 percent of items that provably do not determine an answer, committing to one on the other 97.6
67 percent, or 99.7 percent counting any emitted number. The three earlier checkpoints commit on 99.5
68 percent or more. Naming ##### unanswerable as an available response, and changing nothing else,
69 raises declining to 53.8 percent at the instruct checkpoint. That rate alone is ambiguous, since every
70 probe item is underdetermined and a model offered an escape hatch may simply take it. Running the
71 identical prompt over the untouched parent problems, which are answerable and where declining is
72 an error, gives 1.1 percent. The difference of +52.63 points on [+50.77, +54.57] is the quantity of
73 interest: declining tracks whether the problem can actually be answered.

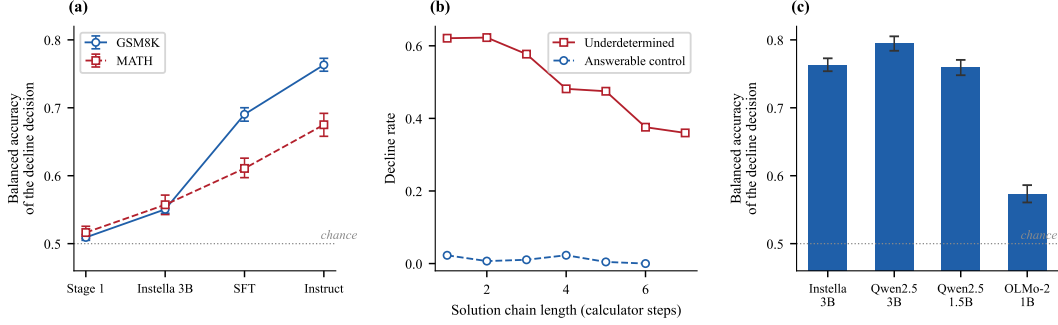


Figure 1: (a) balanced accuracy of the decline decision across the training pipeline, on GSM8K and on an independently built MATH probe, where 0.500 is chance. (b) decline rate against solution chain length at the instruct checkpoint, with the answerable control on the same axes; the control is flat, so the decay is specific to detecting that premises are missing. (c) the same decision quality for instruct checkpoints of three further families.

Table 1 shows the capability developing across the pipeline. Stage one, which precedes the synthetic corpus entirely, discriminates by less than two points. Each subsequent stage improves it, and wrong refusals on answerable problems never exceed 3.5 percent, so the gain does not come from a growing willingness to decline.

A second control rules out the remaining reading, that a pruned problem reads oddly and the model responds to perturbed text. Inserting an irrelevant twenty word sentence carrying an unused number leaves a problem fully answerable while perturbing its surface. Declining moves from 0.0113 clean to 0.0170 with an off topic insertion and 0.0125 with an in domain one, against 0.538 when a premise is actually removed. Rewording the licence while holding its information and its token fixed preserves the ladder, at +0.80, +6.64, +33.41 and +43.77 points with every interval excluding zero, and moves the level: Appendix E reports both.

3.2 Reasoning depth erodes it

The capability is not uniform over problems. Figure 1 plots abstention against the length of the reference solution’s calculator chain. On underdetermined problems at the instruct checkpoint the rate falls from 0.623 at one step to 0.364 at seven, a slope of -5.22 points per step. On the answerable control the slope is -0.11 with an interval containing zero, so the pattern is not a tendency to decline more on short problems. The difference of slopes is -5.11 on $[-6.36, -3.82]$.

The decay steepens as the checkpoint improves, running -2.34 , -3.64 and -5.11 for stage two, supervised fine tuning and instruct. Two alternatives are ruled out. Truncation never exceeds 0.7 percent at any chain length and restricting to terminated generations reproduces the rates exactly. Difficulty is separated by splitting parents on whether the model solves the intact problem, which it does for 76.4 percent. The slope is -4.93 points per step where it can and -4.36 where it cannot, on $[-6.51, -3.36]$ and $[-6.54, -2.19]$. Depth carries information that difficulty does not.

3.3 Both results hold on a second domain and in other families

Two further sweeps address the single benchmark and single family. A premise deletion probe built independently on MATH [Hendrycks et al., 2021], 818 parents and 1091 variants drawn from all seven subjects under the same two validity checks, reproduces the ladder: decision quality runs 0.516, 0.557, 0.611 and 0.675 across the same checkpoints, against 0.509, 0.550, 0.690 and 0.763 on GSM8K. The shape is the same and the level is lower, which is expected, since competition problems are harder and a model declining one it simply cannot solve inflates the answerable rate and shrinks the measured gap. The depth decay reproduces there too, at -6.41 points per additional premise sentence on $[-8.43, -4.49]$, steepening across the pipeline as on GSM8K. The MATH control arm was built without a depth variable, so this replicates the gradient but not its control.

Across families, three further instruct checkpoints graded the same GSM8K items under the same prompt. All four discriminate, with decision quality of 0.763 for Instella 3B Instruct, 0.795 and 0.759 for Qwen2.5 [Qwen Team et al., 2024] at 3B and 1.5B, and 0.573 for OLMo-2 1B Instruct, so premise

110 verification is a property of instruction tuned small language models rather than of one pipeline. The
111 spread is not a ranking, since the models differ in pretraining corpus, instruction tuning recipe and
112 training compute, and nothing in the design attributes a few points to any of those. OLMo-2 is the
113 informative case: its discrimination of +14.68 points on [+12.13, +17.23] clears zero, yet it declines
114 29.2 percent of answerable problems, so its decisions sit near chance. A difference of rates would
115 call that a weaker version of the same capability; the decision metric identifies it as declining freely
116 rather than selectively, which is why false refusal is reported alongside discrimination throughout.

117 4 What exposure returns

118 4.1 Continued pretraining at controlled doses

119 A failure at depth could be a failure to check, or it could be that shallow problems are simply better
120 remembered. Separating those is causal, and observation cannot settle it, so items were injected by
121 continued pretraining at five doses under a fixed 8,388,608 token budget, held constant so a higher
122 dose displaces filler rather than buying compute. Exposure is measured against the 119,014 row
123 subset consumed during stage two rather than the full released pool, which overstates it: 9.65 percent
124 of train parents reach containment 0.999 against the subset and 18.52 percent against the pool. Every
125 arm carries positive controls evaluated before and after: injected document loss must fall and the
126 model must continue an injected document verbatim, since a flat dose response with failed controls
127 would measure a broken training run. Injection worked, verbatim reproduction climbing from 0.133
128 to 0.946 while held out loss rises, so the mixture taught those documents rather than the task.

129 4.2 The chain returns, the answer does not

130 Against a never injected control at the same dose, reproduction of the memorised calculation chain
131 reaches +5.44 points on [+1.46, +9.38] at 64× and rises with dose, while reproduction of the final
132 answer reaches +1.52 on [−2.12, +5.21], an interval containing zero. The model reproduces the
133 source text almost exactly and still does not recover the deleted quantity or the answer it determines.
134 Repeating the identical mixture, budget, seed and probe on OLMo-2-1B, Qwen2.5-1.5B and Qwen2.5-
135 3B gives +0.28, +0.00 and +2.28 points at the top dose, every interval containing zero, across ten
136 dose points. Qwen2.5-3B is the sharpest case, driven to verbatim reproduction of 1.000 and still
137 recovering the deleted answer no more often than a model that never saw the documents. Chain
138 reproduction is scored on three of the four models. It holds on Instella and is underpowered on the
139 other two, as Appendix D sets out.

140 5 Discussion

141 Two things a single accuracy number conflates come apart here. Producing an answer is something
142 every checkpoint does readily, including on 97.6 percent of problems that have none. Verifying that
143 the premises support one appears only after post training, and then degrades with the depth of the
144 chain being tracked. Benchmark scoring cannot see the difference, because every item it scores is
145 answerable. Reporting declining under a prompt that permits it, against an answerable control, costs
146 one evaluation pass and makes the second capability visible.

147 **Limitations:** One problem generator for the injection arm. Detection is measured through an
148 explicit abstention option, so a model that recognises insufficiency but lacks the vocabulary to say so
149 is scored as failing. Difficulty is stratified by the model’s own accuracy rather than an independent
150 measure, and depth on MATH is sentence count, since that benchmark annotates no calculation chain.
151 Chain reproduction under injection is established on one model, and the replication is underpowered
152 for it. Attribution to specific training documents is not established anywhere.

153 **Conclusion:** Measured against containment rather than an estimate, Instella’s grade school arith-
154 metic is not recovered text: no test item approaches containment against the corpus, and heavy
155 injection returns the memorised chain without the answer. What the probe reveals instead is a
156 separable capability. The model detects that a problem no longer determines an answer, does so
157 increasingly well across training, and loses that detection as the chain lengthens.

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A Related work

Premise verification and abstention: Declining to answer became a measured quantity with SQuAD 2.0, which added fifty thousand adversarially written unanswerable questions and scored systems on abstaining [Rajpurkar et al., 2018]. That models fail at the mathematical version of this is well documented. QuestBench formalises underspecified reasoning as a constraint satisfaction problem missing one variable and finds that models fail to identify the missing variable even where they solve the fully specified problem [Li et al., 2025a]. AbstentionBench gathers unanswerable questions across domains and reports that reasoning fine tuning degrades abstention rather than improving it [Kirichenko et al., 2025]. PCBench evaluates premise critique over fifteen models and finds that most detect a flawed premise only under explicit prompting, which is the same licensing dependence measured in Section 3.1 [Li et al., 2025b], and ReliableMath establishes the corresponding benchmark for unsolvable mathematics [Xue et al., 2025]. Concurrent work sharpens two pieces of this. Gu et al. [Gu et al., 2026] name the gap between detecting insufficiency and acting on it and train an explicit answerability decision to close it; Wagner [Wagner, 2026] separates confidence in an answer from confidence that a question is answerable and shows the usual confidence signal is nearly blind to the second. The present work differs in what it varies. Those studies evaluate released models, so they can establish that the capability is uneven but not where it comes from or what removes it. Measuring four checkpoints of one published pipeline turns the capability into a developmental quantity, and reports two gradients rather than a level: it is manufactured almost entirely by post training, and it decays with the depth of the reasoning chain, faster in the checkpoints that verify best. The AbstentionBench finding that reasoning fine tuning degrades abstention and the ladder reported here are not in conflict on this reading, since the depth gradient is a within checkpoint effect that a single benchmark average over problems of mixed depth would partially absorb.

Memorisation and contamination: Huang et al. [Huang et al., 2025] construct hand authored hard perturbations of competition mathematics and attribute accuracy loss under an unchanged solution method to blind application of a learned procedure, a reading broadly consistent with the procedure level memorization in Section 4.2, established here against a corpus verified exactly rather than inferred. Wu et al. [Wu et al., 2025] show that reinforcement learning gains on mathematical reasoning depend on the pretraining exposure of the base policy and argue that clean synthetic problems are needed for reliable conclusions, a concern the checkpoint based design addresses by evaluating a policy before any relevant exposure occurred. Zhang et al. [Zhang et al., 2024] rebuild GSM8K from a contamination free procedure and report accuracy gaps of up to 13 points that track prior exposure across closed models, using an estimated exposure signal in the absence of any released training corpus; the present work substitutes an exact containment measurement for that estimate on a family where the corpus is public.

B Containment method, calibration, and corpus routes

The within split contamination contrast, high against low containment GSM8K train items, was closed rather than merely complicated by two findings. The corpus subset actually consumed during Instella 3B stage two pretraining, `train_119K`, is a 119,014 row subset of a 1,367,882 row released pool; containment computed against the pool over counts exposure, since 12.57 percent of GSM8K train reaches 0.80 or higher against the subset actually consumed against 23.34 percent against the pool, both measured on the same 891 parent problems through one extraction path. No GSM8K test item reaches containment 0.50 against either corpus, and the maximum over the full pool is 0.487, so the released corpus is train derived and the split remains a usable exposure contrast. Independently, `allenai/tulu-3-sft-mixture` is itself part of the Instella stage two pretraining mixture and `nvidia/OpenMathInstruct-2` enters at supervised fine tuning; both are generated in part from GSM8K train, and both carry it at containment 0.999 or higher on more than 97 percent of items. Every GSM8K train item therefore entered post stage one training through at least one of these routes regardless of its containment in the Instella specific synthetic corpus, so a high against low contrast defined on that corpus alone would compare two groups that are both almost entirely exposed once these mixtures are accounted for; the contrast is not attenuated by this, it is unidentifiable.

Item text is normalized to lowercase alphanumeric tokens and decomposed into every window of 13 consecutive tokens, following the overlap convention of Brown et al. [Brown et al., 2020]. Containment against a document is the fraction of an item’s windows found in that document;

containment against a corpus is the maximum of that fraction over every document, computed in one streaming pass with an inverted index from window to source item so cost scales linearly in corpus size. Because the corpus is generated only from the GSM8K training split, every GSM8K test item is a verified negative, and the false positive rate is measured rather than assumed: at the adopted threshold of 0.80, zero of 1319 test items are flagged, a Wilson 95 percent upper bound of 0.29 percent; a looser rule flagging any window match flags 3 of 1000 sampled test items, 0.30 percent.

Within the Tulu 3 mixture, a sampled prefix of 300,000 rows, six of the mixture’s subsets, finds GSM8K train items reaching containment 0.999 through the `flan_v2_converted` subset, 6021 of 7473 items from 89,982 rows of that source, and through `wildchat` conversation logs, 4 items from 100,000 rows, an organic route in which users pasted benchmark problems into chat. Every scan reports rows examined alongside items found, since a sampled prefix bounds containment from below and a zero means absent from the rows scanned rather than absent from the corpus.

C The irrelevance marker

The inserted frame in the crossed design ends "and has no bearing on this question", which is an instruction to ignore the insertion carried inside the insertion itself. Shi et al. [Shi et al., 2023] report that such an instruction mitigates the loss irrelevant context causes, so the modest cost measured in Section 3.3 may sit inside their mitigated regime rather than in tension with it. Testing that requires removing the clause and nothing else. The replacement, "on the afternoon of the same day", holds the sentence at exactly twenty words and preserves position, topic noun and unused integer.

Table 2: Accuracy against each problem’s own original, and the effect of removing the irrelevance marker. Cluster bootstrap over parent problems, 4000 draws, 1764 parents.

Checkpoint	Marked	Unmarked	Marker effect
Instella 3B	−2.21 [−3.88, −0.51]	−5.67 [−7.43, −3.77]	−3.46 [−4.90, −2.01]
SFT	+0.09 [−1.53, +1.67]	+0.11 [−1.53, +1.81]	+0.03 [−1.33, +1.36]
Instruct	−2.21 [−3.71, −0.65]	−2.78 [−4.34, −1.10]	−0.57 [−1.93, +0.79]

The marker governs the cost at the base stage two checkpoint and not after post training. One checkpoint of three is a single result rather than a pattern, and it is reported as such: it accounts for the size of the stage two effect and does not account for the residual two points at Instruct, which neither the number, nor topicality, nor the marker explains.

D The two withdrawn checks, and the dose response in full

The clause ladder is an externally authored perturbation suite whose rungs share template identifiers, which invites pairing each perturbed item against its own template head. The instances at each rung are independently generated rather than derived from one another, so a later rung is a different and harder problem set rather than the same problems with clauses appended, and an accuracy difference across rungs therefore mixes added content with added difficulty. Premise reordering permutes the non question sentences of a problem and preserves the gold answer by construction, but a narrative word problem often carries pronouns, temporal connectives and back references across sentences, so a permutation can leave text that no reader could solve. Filtering to items where no premise after the first carries such a dependency retains 24 percent of the set, and on that subset the effect is +2.2 points on [−6.5, +10.9] over 46 problems. Manual inspection finds violations that survive the filter, so even that residual is a spot check rather than an estimate.

Table 3 gives the dose response in full. Two outcomes are scored on the same generations. Answer recall is reproduction of the deleted parent’s original answer; procedure reproduction is the fraction of the parent’s intermediate calculator results appearing in the generation, excluding the final answer. Both are differences in differences, injected minus held out at a dose, minus the same quantity at dose zero, so the 8,388,608 tokens every arm trains on cancel. The dose zero row is the reference rather than a result, and its positive controls fail by construction, since a model that was never shown the documents cannot have memorised them.

Answer recall is flat in every model at every dose, including the two arms driven to near perfect verbatim reproduction, which is the result the body reports. Procedure reproduction rises with dose

Table 3: Continued pretraining at five doses on four base models. Verbatim is the fraction of injected documents the model continues exactly after training. Answer and procedure are differences in differences against the dose zero arm, in percentage points, with cluster bootstrap intervals over parent problems, 4000 draws, seed 6198. Only one cell excludes zero, marked *. Qwen2.5-3B was run at two doses and is not scored for procedure reproduction.

Model	Dose	Verbatim	Answer recall	Procedure reproduction
Instella 3B	0×	0.133	reference	reference
	1×	0.153	+0.06 [−2.99, +3.17]	+0.34 [−2.86, +3.67]
	4×	0.180	+0.09 [−2.48, +2.68]	+0.73 [−2.27, +3.90]
	16×	0.491	−0.22 [−3.56, +3.10]	+2.78 [−0.79, +6.42]
	64×	0.946	+1.52 [−2.12, +5.21]	+5.44 [+1.46, +9.38]*
Qwen2.5 3B	0×	0.277	reference	not scored
	16×	0.713	+1.41 [−1.71, +4.60]	not scored
	64×	1.000	+2.28 [−1.21, +5.77]	not scored
Qwen2.5 1.5B	0×	0.243	reference	reference
	1×	0.250	+0.28 [−1.99, +2.60]	−2.17 [−5.05, +0.65]
	4×	0.255	+1.95 [−0.85, +4.84]	−0.70 [−3.69, +2.32]
	16×	0.406	+0.84 [−2.13, +3.71]	+0.83 [−2.65, +4.38]
	64×	0.989	+0.00 [−2.46, +2.52]	+0.33 [−3.49, +4.16]
OLMo 2 1B	0×	0.126	reference	reference
	1×	0.128	−0.32 [−2.38, +1.67]	−1.40 [−3.73, +0.93]
	4×	0.172	+0.00 [−1.90, +1.77]	−0.52 [−3.32, +2.30]
	16×	0.207	+0.62 [−1.56, +2.83]	+0.26 [−3.02, +3.49]
	64×	0.727	+0.28 [−2.34, +2.73]	−0.30 [−4.29, +3.61]

on Instella and reaches an interval excluding zero at the top dose only. On Qwen2.5 1.5B and OLMo 2 1B it does not, and the point estimates there are not consistently positive, so the chain result should be read as established on one model rather than replicated. The two replication models also memorise less at the same dose, reaching verbatim 0.989 and 0.727 against Instella’s 0.946 at 64× but only 0.406 and 0.207 at 16×, so the doses at which the Instella effect appears are not matched in memorisation strength. Distinguishing a genuine family difference from insufficient memorisation would need doses set to match verbatim reproduction rather than repetition count, which is not what these runs hold fixed.

E Robustness of the licence to its wording

The result in Section 3.1 is produced by one sentence, so it is worth knowing how much of it belongs to that sentence. The licence was reworded to carry the same three instructions, work stepwise, mark the final answer with the same token, and decline with ##### unanswerable when the problem does not fix a unique answer, in different words. The abstention token is held fixed so the scorer is unchanged between conditions and any movement is attributable to wording rather than to what is counted. All 3179 pruned items and all 1779 answerable parents were regenerated on all four checkpoints.

Table 4: The developmental ladder under two wordings of the same licence. Discrimination is declining on underdetermined items minus declining on answerable ones, in percentage points; false decline is the rate on answerable parents. Cluster bootstrap over parent problems, 4000 draws, seed 6198. Every interval excludes zero in both conditions.

Checkpoint	Discrimination		Decision quality		False decline	
	Original	Reworded	Original	Reworded	Original	Reworded
Stage 1	+1.84	+0.80	0.509	0.504	0.025	0.007
Instella 3B	+10.10	+6.64	0.550	0.533	0.008	0.008
SFT	+38.08	+33.41	0.690	0.667	0.035	0.008
Instruct	+52.63	+43.77	0.763	0.719	0.011	0.006

Two things separate. The developmental claim is robust to the wording: the ladder is monotone under both licences, in the same order, with every interval excluding zero, and false declines stay below one percent throughout the reworded condition. The level is not. Discrimination at instruct falls from +52.63 to +43.77 points and declining on underdetermined items falls from 0.538 to 0.444, so the

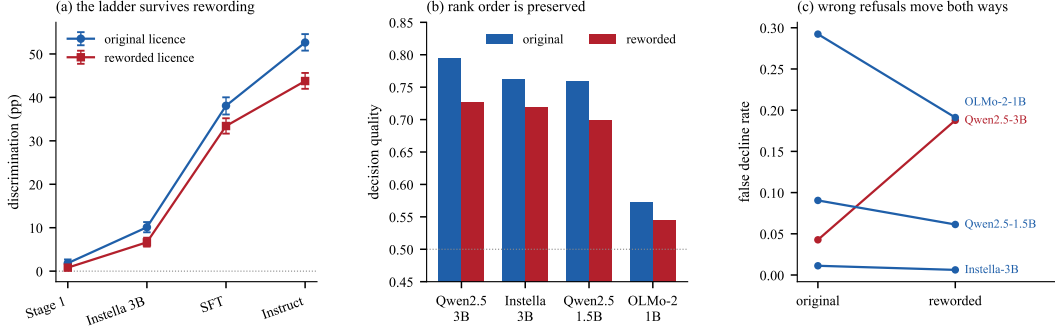


Figure 2: Robustness of the result to the wording of the licence. (a) the developmental ladder under both wordings, with cluster bootstrap intervals; the shape and ordering are preserved and every interval excludes zero, while the level falls. (b) decision quality across families on identical items, where the rank order is unchanged. (c) the false decline rate, which is the quantity that actually moves, and which moves in opposite directions across models: Qwen2.5 3B more than quadruples while OLMo 2 falls and Instella stays near zero. Drawn from the same analysis files as Tables 4 and 5.

326 rates quoted in Section 3.1 should be read as specific to the licence that produced them rather than
 327 as properties of the checkpoint. The gap between conditions widens as the checkpoint improves,
 328 running -1.03 , -3.46 , -4.66 and -8.87 points, so the checkpoints that verify best are also the most
 329 sensitive to how they are asked. Truncation does not explain the shift, being equal or lower under the
 330 reworded licence at every checkpoint, and markedly lower at stage one, at 0.133 against 0.211 on
 331 pruned items.

332 The same rewording was applied to the cross family sweep of Section 3.3, on the identical items
 333 (Table 5). All four instruct checkpoints still discriminate with every interval excluding zero, and the
 334 ordering by decision quality is unchanged, so the comparison in Section 3.3 does not depend on the
 335 licence used to obtain it. Levels fall under rewording here as well. The false decline rate is the cell
 336 that moves most and it does not move uniformly: Qwen2.5 3B rises from 0.043 to 0.188 and OLMo 2
 337 falls from 0.292 to 0.191 , while Instella stays near zero at 0.006 . The spread in how models respond
 338 to the wording of a licence is therefore wider than the spread in how well they discriminate, which is
 339 a further reason to report false refusal separately rather than folding it into a single rate.

Table 5: The cross family sweep under both wordings, identical GSM8K items. Every interval excludes zero in both conditions and the ordering by decision quality is preserved.

Instruct checkpoint	Discrimination		Decision quality		False decline	
	Original	Reworded	Original	Reworded	Original	Reworded
Qwen2.5 3B	+58.92	+45.33	0.795	0.727	0.043	0.188
Instella 3B	+52.63	+43.77	0.763	0.719	0.011	0.006
Qwen2.5 1.5B	+51.88	+39.83	0.759	0.699	0.090	0.061
OLMo 2 1B	+14.68	+9.17	0.573	0.546	0.292	0.191

340 F Reproduction

341 All generation, containment, and analysis code, together with every scored generation file and
 342 containment record underlying this work, is released for audit at <https://anonymous.4open.science/r/Instella-Reasoning-5F18>, with a paired dataset mirror holding the generation files.
 343 Both are anonymized for review and will be de anonymized on acceptance. Every containment number
 344 reported uses the maximum over individual documents definition stated above; an earlier internal
 345 computation of the same quantity against the corpus pooled as one text produced a materially different
 346 figure for the same released split and is superseded by the document level definition throughout this
 347 work. Random seeds are fixed across item selection, deletion variant construction, the continued
 348 pretraining mixture, and every cluster bootstrap. An assistant was used to help implement generation,
 349 containment, and analysis code under the direction of the authors, who independently verified every
 350 reported number against the underlying generation files; the assistant did not design the probes or
 351 select which results to report.
 352

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510 Answer: [Yes]

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